

Course Outline

EN.705.743: 🧠 ChatGPT from Scratch: Building and Training Large Language Models **Fall 2026**

This outline provides an overview of the course and assignments by week. Please also check the Calendar in Canvas for specific due dates. Each course module runs for seven (7) days, i.e., one week, starting on the day of the first week (**Tuesdays**). Due dates for assignments are referred to by the day of the module week in which they are due. For example, if an Assignment is due by Day 7, then the Assignment should be completed by **Monday** or the 7th day of the module.

Module	Dates	Module Title	Assignments
Module 1	Tue 09/01	Introduction to ChatGPT and LLMs	Assignment 1 due
Module 2	Tue 09/08	Tokenization & Embeddings	Assignment 2 due
Module 3	Tue 09/15	Multi-Head Attention	Assignment 3 due
Module 4	Tue 09/22	Transformers	Assignment 4 due
Module 5	Tue 09/29	BERT and Encoder Models	- Assignment 5 due Project Topic Proposal due
Module 6	Tue 10/06	Training LLMs	Assignment 6 due
Module 7	Tue 10/13	Sampling & Inference (with Practical Inference)	Assignment 7 due
Module 8	Tue 10/20	Fine-Tuning	Assignment 8 due
Module 9	Tue 10/27	Parameter Efficient Tuning	Assignment 9 due
Module 10	Tue 11/03	Quantization	Assignment 10 due
Module 11	Tue 11/10	RAG And Agents	Assignment 11 due
Module 12	Tue 11/17	Multimodality (Extending LLMs beyond text)	Assignment 12 due
11/23–11/29 - Thanksgiving break - Nothing assigned or due			
Module 13	Tue 12/01	Reinforcement Learning with Human Feedback (Aligning LLMs to human values)	Project Presentation due
Module 14	Tue 12/08	Scaling Laws, Foundation Models	Research Paper due

